Full Stack Assignment – 07

Assignment ID: FSD-07

Title: Event Registration Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop an **Event Registration Management System** that allows users to register for events while enabling organizers to manage registrations.
The system must enforce capacity constraints and ensure **data persistence using a database**.

2️⃣ User Roles

USER

- Register and log in to the system
- View available events
- Register for an event
- View own registration status

ORGANIZER

- Log in to the system
- Create and manage events
- View all event registrations

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Event Management

- ORGANIZER can create events with capacity limits
- Event details must be stored in the database
- Event availability must be fetched dynamically

Event Registration

- USER can register for available events
- Registration details must be stored in the database

- USER should be able to view registration status

4️⃣ Business Rule (Critical)

⚠️ **Event registration must close automatically when the maximum capacity is reached.**

Once full, no further registrations must be allowed.

5️⃣ Technical Constraints (Mandatory)

- All event and registration data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Environment variables must be used for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and ORGANIZER
- Event listing with available capacity
- Event registration action for USER
- Registration overview for ORGANIZER

UI design is not evaluated; logic and correctness matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used

- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

The assessment emphasizes **constraint handling, role-based logic, and end-to-end execution**.